1. Let r be the distance

$$\sqrt{x^2 + y^2 + (z-2)^2} = r$$

$$\boxed{x^2 + y^2 + (z-2)^2 = r^2}$$

2. Let (x, y, z) be point.

Distance to $z = -1$: $z - (-1) = z + 1$

Distance to $(0, 0, 1)$: $\sqrt{x^2 + y^2 + (z-1)^2}$

$$\sqrt{x^2 + y^2 + (z-1)^2} = z + 1$$

$$x^2 + y^2 + z^2 - 2z + 1 = z^2 + 2z + 1$$

$$6\left(z + \frac{1}{2}\right) = x^2 + y^2$$

Paraboloid

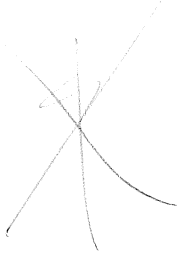
$$z = \frac{x^2 + y^2}{6} - \frac{1}{2}$$

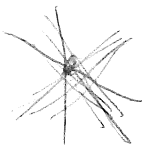
$$\boxed{x^2 + y^2 = 4z}$$

3a) trace $x=0$: $y^2 - z^2 = 0 \Rightarrow y = \pm z$

trace $y=0$: $x^2 - z^2 = 0 \Rightarrow x = \pm z$

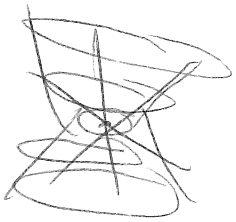
trace $z=0$: $x^2 + y^2 = 0 \rightarrow$ ~~$x^2 = -y^2$~~ $x=0, y=0$





$x=3$: $9 + y^2 - z^2 = 0$

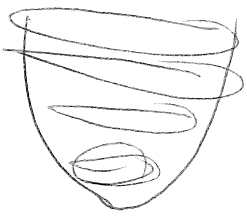
$z^2 - y^2 = 9$

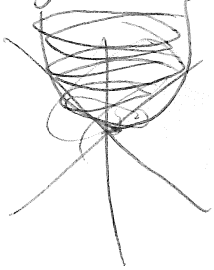


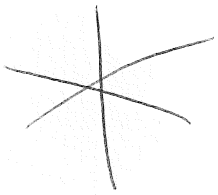
Cones

b) $x=0$: $y^2 = z$

$y=0$: $x^2 = z$ Cute







c) $x^2 - y^2 - z = 0$

$x=0$: $-y^2 - z = 0$

~~$y^2 = -z$~~

$y=0$: $x^2 - z = 0$

$z = x^2$

$z=0$: $x^2 - y^2 = 0$

$x = \pm y$

$x^2 - y^2 = z$

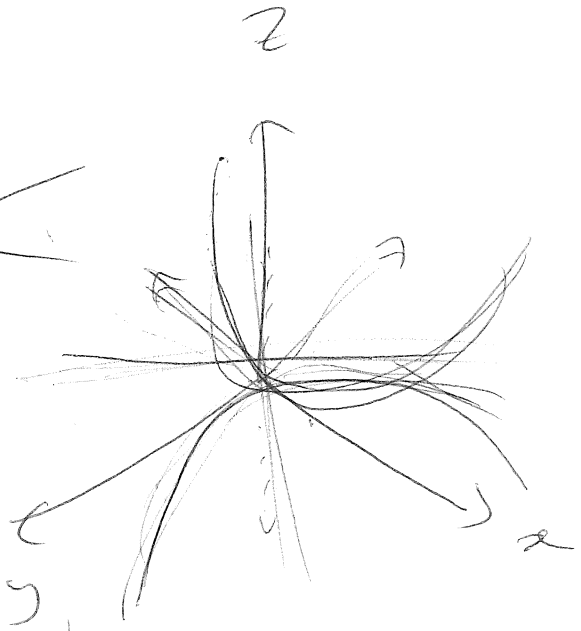
$z>0$: $x^2 - y^2 = (\sqrt{z})^2$

$z<0$: $x^2 - y^2 = -|z|$

$|z| = y^2 - x^2$

$(\sqrt{|z|})^2 = y^2 - x^2$







d) $x^2+y^2+z^2=4$

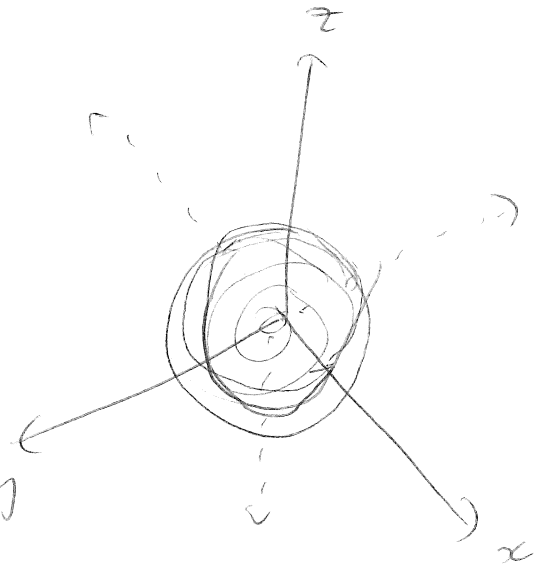
$x=0: \; y^2+z^2=4$

$y=0: \; x^2+z^2=4$

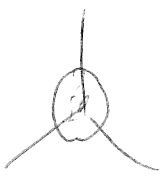
$z=0: \; x^2+y^2=4$

$x^2+y^2=4-z^2$

$\sqrt{4-z^2}$ radius

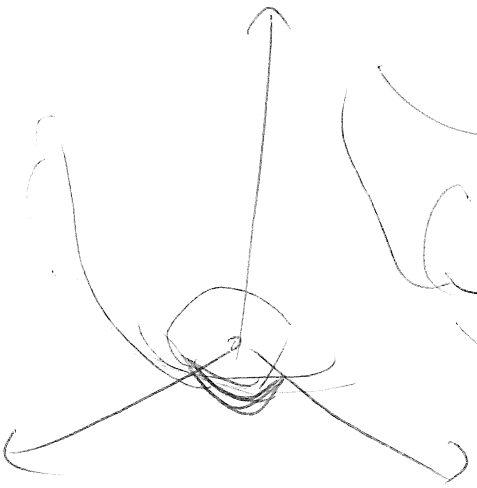


Sphere



e) $x^2+y^2-z^2=4$

$x^2+y^2=4+z^2$

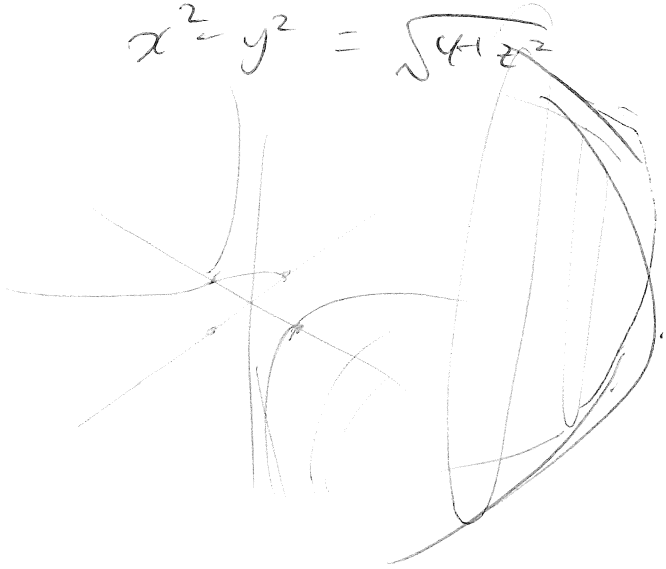




f) $x^2-y^2-z^2=4$

$x^2-y^2=4+z^2$

$x^2-y^2=\sqrt{4+z^2}$



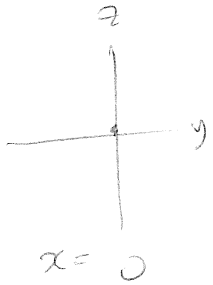


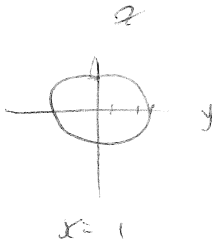
4. $y^2 + z^2 = (4x - x^2)^2$

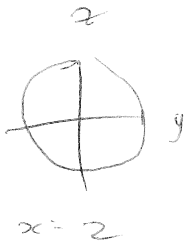
$x = a$: $y^2 + z^2 = (4a - a^2)^2$

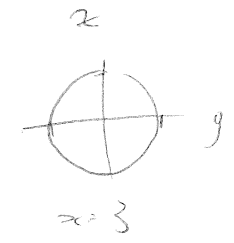
$= a^2(4-a)^2$

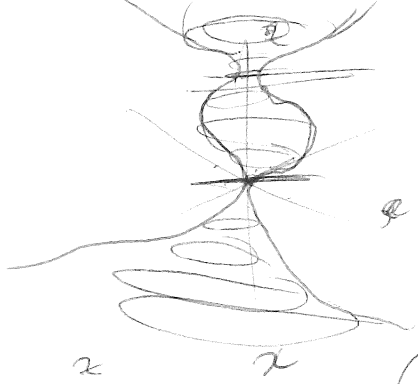
circle











$(4x - x^2)^2$

$(x(4-x))^2$

$x^2(4-x)^2$

